Lecture 12: An Example on Step Response to RLC Series Circuit

In the previous class:

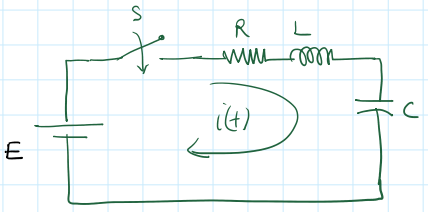
$$Ri + L\frac{di}{dt} + \frac{1}{C}\int_{-\infty}^{t} i\,dt = E$$

$$\Downarrow$$

$$\frac{d^2i}{dt^2} + \frac{R}{L}\frac{di}{dt} + \frac{i}{LC} = \frac{E - V_{C0}}{L}$$

$$\Downarrow$$

$$i(t) = \frac{E - V_{C0}}{\omega_d L} e^{-\zeta\omega_n t} \sin\omega_d t = \frac{E - V_{C0}}{\sqrt{1-\zeta^2}} 2\zeta e^{-\zeta\omega_n t} \sin\omega_d t$$



where $\omega_d = \omega_n\sqrt{1-\zeta^2}$

$$\omega_n = \sqrt{\frac{1}{LC}}\ ;\quad R_c = 2\sqrt{\frac{L}{C}}\ ;\quad \zeta = \frac{R}{R_c} = \frac{R}{2}\sqrt{\frac{C}{L}}$$

$$\frac{R}{L} = 2\zeta\omega_n$$

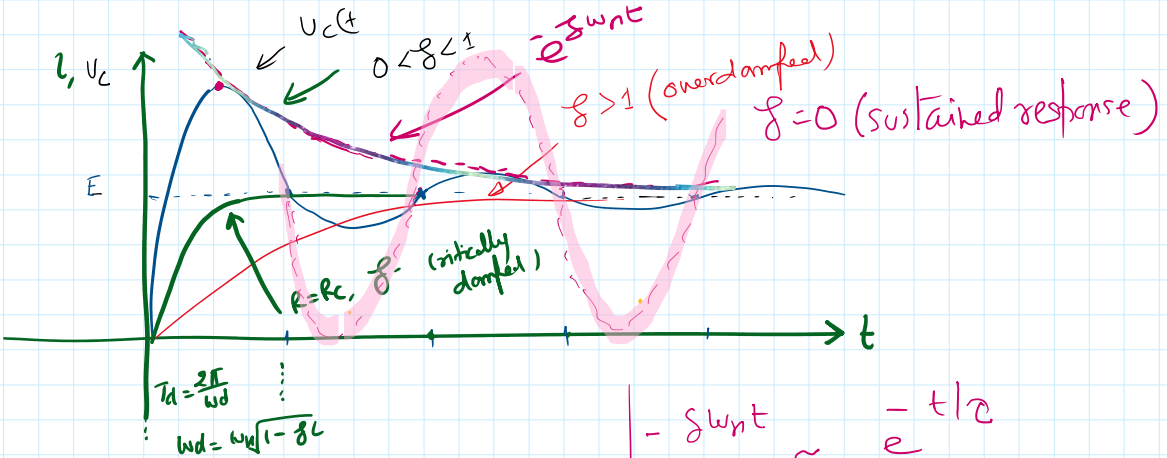
With these we established expressions for voltages,

$$v_R(t) = \frac{E - V_{C0}}{\sqrt{1-\zeta^2}}\ 2\zeta e^{-\zeta\omega_n t} \sin\omega_d t$$

$$v_L(t) = L\frac{di}{dt} = -\frac{E - V_{C0}}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t} \sin(\omega_d t - \phi)\ ;\quad \phi = \tan^{-1}\frac{\sqrt{1-\zeta^2}}{\zeta}$$

$$v_C(t) = \frac{1}{C}\int i\,dt = E - v_R - v_L = E - \frac{E - V_{C0}}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t} \sin(\omega_d t + \phi)$$

$$= E\left\{1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}} \sin(\omega_d t + \phi)\right\}$$



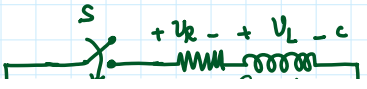
$$e^{-\zeta\omega_n t} \simeq e^{-t/\tau}$$

For $\tau = 1/\zeta\omega_n$

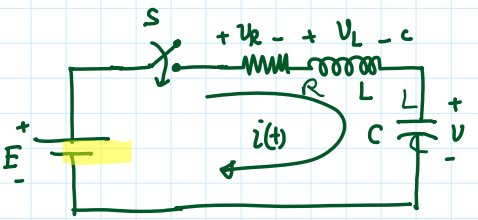
Here time constant $\tau = \frac{1}{\zeta\omega_n}$, thus

$$v_C(t) = E\left[1 - \frac{e^{-t/\tau}}{\sqrt{1-\zeta^2}} \cdot \sin(\omega_d t + \phi)\right] \text{ for } \phi = \tan^{-1}\frac{\sqrt{1-\zeta^2}}{\zeta}$$

Example: Consider RLC series circuit



Example: Consider RLC series circuit supplied by battery of E volts. The switch S is closed at t=0. The initial conditions are zero. Determine the following.



(a) Characteristic equation

(b) Damping factor, ζ, critical resistance, R_c, natural frequency, ω_n, damped frequency, ω_d.

(c) Determine current response i(t)

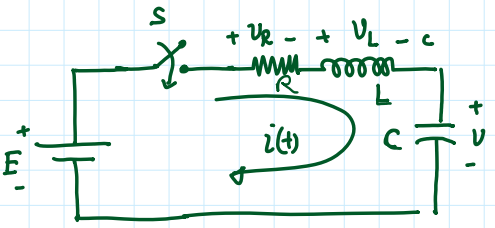
(d) The voltages across R, L and C

(e) For what value of R, the system response is overdamped.

① Overdamped $\zeta > 1$; $R > R_c$
② Critically damped $\zeta = 1$, $R = R_c$
③ Underdamped $0 < \zeta < 1$ $0 < R < R_c$
④ Sustained oscill: $\zeta = 0$, $R = 0$

$E = 1000\,V$, $R = 1000\,\Omega$, $L = 1\,H$, $C = 1\,\mu F$

$V_{C0} = 0$, $i_{L0} = 0$



(a) $Ri + L\frac{di}{dt} + \frac{1}{C}\int i\,dt = E$

$\downarrow$

$\frac{d^2i}{dt^2} + \frac{R}{L}\frac{di}{dt} + \frac{i}{LC} = 0$

$\downarrow$

$s^2 + \frac{R}{L}s + \frac{1}{LC} = 0$

$\downarrow$

$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$

$s^2 + 1000s + 10^6 = 0$

$\omega_n = \sqrt{\frac{1}{LC}} = \frac{1}{\sqrt{1\times10^{-6}}} = 10^3$ rad/sec

$\frac{R}{L} = \frac{1000}{1} = 1000$

$\zeta = \frac{R}{R_c} = \frac{R}{R_c}$ ← $\left|\ \frac{R_c}{2L} = \frac{1}{\sqrt{LC}}\right.$

$R_c =$

$\zeta > 1$ (overdamped)
$0 < \zeta < 1$ underdamped
$\zeta = 0$ Sustained oscillation
$\zeta = 1$ Critical damping

$\omega_d = \sqrt{1-\zeta^2}\ * \omega_n$

$= 1000\sqrt{1-0.5^2}$

$= 1000 \times \frac{\sqrt{3}}{2} = 0.866 \times 1000$

$\boxed{\omega_d = 866 \text{ rad/sec}}$

$R > R_c$ (case 1), overdamped.

$i(t)$

$\frac{(E-V_0)}{\omega_d L}\, e^{-\zeta\omega_n t} \sin\omega_d t$

$i(t) = \frac{E}{\omega_d L}\, e^{-\zeta\omega_n t} \sin\omega_d t$

$= \frac{1000}{866 \times 1}\, e^{-0.5\times1000\, t} \sin 866\, t$

$2\sqrt{\frac{L}{C}} = 2000\,\Omega$

$= \frac{1000}{2000} = 0.5$

$\boxed{\zeta = 0.5}$

$$V_R(t) = iR = \frac{1000}{866.67 \times 1} e^{-500t} \sin 866t$$

$$\boxed{i(t) = 1.154\, e^{-500t} \sin 866t}$$

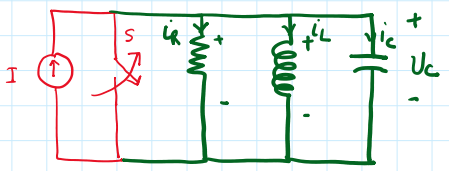
Now you can find $V_L = -\frac{E - V_{C0}}{\sqrt{1-\delta^2}} e^{-\delta\omega_n t} \sin(\omega_d t - \phi)$

$$\boxed{V_L = -1154\, e^{-500t} * \sin(866t - \pi/3)}$$

$$\phi = \tan^{-1}\frac{\sqrt{1-(0.5)^2}}{0.5} = 60^\circ$$

and $$\boxed{V_C = E\,(1 - 1.154\, e^{-500t} \sin(866.67t + \pi/3)}$$

RLC Parallel Circuit



$$I = i_R + i_L + i_C$$

$$\downarrow$$

$$\frac{V_C}{R} + \frac{1}{L}\int_{-\infty}^{t} V_C\,dt + C\frac{dV_C}{dt} = I \qquad V_C = L\frac{di_L}{dt} \qquad V_C = V_L = V_R$$

$$\downarrow$$

$$\frac{V_C}{R} + \frac{1}{L}\int_{-\infty}^{0} V_C\,dt + \frac{1}{L}\int_{0}^{t} V_C\,dt + C\frac{dV_C}{dt} = I$$

$$\frac{V_C}{R} + i_{L0} + \frac{1}{L}\int_{0}^{t} V_C\,dt + C\frac{dV_C}{dt} = I$$

$$\frac{V_C}{R} + \frac{1}{L}\int_{0}^{t} V_C\,dt + C\frac{dV_C}{dt} = (I - i_{L0})$$

Differentiating it results to,

$$C\frac{d^2V_C}{dt^2} + \frac{1}{R}\frac{dV_C}{dt} + \frac{V_C}{L} = 0 \qquad i_C = C\frac{dV_C}{dt}$$

$$\boxed{\frac{d^2V_C}{dt^2} + \frac{1}{RC}\frac{dV_C}{dt} + \frac{V_C}{LC} = 0}$$

Which is of the form: $\frac{d^2V_C}{dt^2} + a\frac{dV_C}{dt} + bV_C = C$
and can be solved in a similar way, with following change:

$$V_C \leftarrow i$$

$$\frac{1}{RC} \leftarrow \frac{R}{L}$$

This time the roots of characteristic equation are

$$S_{1,2} = \frac{-\frac{1}{RC} \pm \sqrt{\left(\frac{1}{RC}\right)^2 - 4/LC}}{2} = -\frac{1}{2RC} \pm \sqrt{\left(\frac{1}{2RC}\right)^2 - \frac{1}{LC}}$$

$$s_{1,2} = \frac{-\frac{1}{RC} \pm \sqrt{\left(\frac{1}{RC}\right)^2 - 4/LC}}{2} = -\frac{1}{2RC} \pm \sqrt{\left(\frac{1}{2RC}\right)^2 - \frac{1}{LC}}$$

Similar to the previous, there will be four cases:

(A) overdamped: $\left(\frac{1}{2RC}\right)^2 > \frac{1}{LC}$, two distinct real roots

(B) Critically damped: $\left(\frac{1}{2RC}\right)^2 = \frac{1}{LC}$; two equal roots

(C) Underdamped: $\left(\frac{1}{2RC}\right)^2 < \frac{1}{LC}$; complex conjugate roots

(D) Sustained oscillation: $R = 0$; pure imaginary roots.

Similar to the previous: Now

1. Natural frequency: $\omega_n = \sqrt{\frac{1}{LC}}$
2. Critical Resistance: $\frac{1}{2R_C} = \sqrt{\frac{1}{LC}}$

$$R_c = \frac{1}{2C} * \sqrt{LC} = \frac{1}{2}\sqrt{\frac{L}{C}}$$

3. Damping ratio: $\zeta = \frac{R_c}{R}$ (reverse ratio to that of series RLC circuit)

$$\zeta = \frac{1}{2R}\sqrt{\frac{L}{C}}$$

$$\Rightarrow \quad = \frac{1}{2R_C} * C * \sqrt{\frac{L}{C}} = \frac{1}{2RC} \cdot \sqrt{LC} * \frac{1}{}$$

$$\zeta = \frac{1}{2RC} * \frac{1}{\omega_n}$$

$$\Rightarrow \boxed{\frac{1}{RC} = 2\zeta\omega_n}$$

Thus, Characteristic equation:

$s^2 + \frac{s}{RC} + \frac{1}{LC} = 0$ become

$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$ (exactly similar to the previous case

where $\zeta\omega_n = \frac{1}{2RC}$; $\omega_n = \frac{1}{\sqrt{LC}}$

Hence the solution can be written as,

$$v_c(t) = k_1 e^{s_1 t} + k_2 e^{s_2 t}$$

$$= e^{-\zeta\omega_n t}(k_1' \cos\omega_d t + k_2' \sin\omega_d t)$$

Applying initial condition: At $t = 0$, $v_c(0) = 0$

$$\Rightarrow k_1' = 0$$

Thus $v_c(t) = k_2' e^{-\zeta\omega_n t} \sin\omega_d t$

k_2' is found by applying initial condition $\left.\frac{dv_c}{dt}\right|_{t=0}$:

$$\frac{dv_c}{dt} = k_2' e^{-\zeta\omega_n t}[-\zeta\omega_n \sin\omega_d t + \omega_d \cos\omega_d t]$$

At $t = 0$; $\frac{v_c}{R} + \frac{1}{L}\int_0^t v_c dt + C\frac{dv_c}{dt} = (I - i_{L0})$

At $t = 0$ $\frac{dv_c}{dt} = \frac{(I - i_{L0})}{C}$

$$\Rightarrow \frac{(I - i_{L0})}{C} = k_2' * \omega_d$$

$$\Rightarrow k_2' = \frac{(I - i_{L0})}{\omega_d C}$$

$$\Rightarrow \quad K_2' = \frac{(I - i_{L0})}{\omega_d C}$$

Thus

$$\boxed{v_C(t) = \frac{I - i_{L0}}{\omega_d C} e^{-\zeta\omega_n t} \sin \omega_d t}$$

Once we know the voltage, we can find out i_R, i_C and i_L as follows.

$$i_R = \frac{v_C}{R} = \frac{I - i_{L0}}{\omega_d}\left(\frac{1}{RC}\right) \cdot e^{-\zeta\omega_n t} \sin \omega_d t$$

$$= \frac{I - i_{L0}}{\omega_n \sqrt{1-\zeta^2}} \cdot 2\zeta\omega_n e^{-\zeta\omega_n t} \sin \omega_d t$$

$$i_R(t) = \frac{(I - i_{L0})}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t} \; 2\zeta \sin \omega_d t \quad \text{---} \quad (*)$$

$$i_C(t) = C\frac{dv_C}{dt} = C * \frac{(I - i_{L0})}{\omega_d C} \frac{d}{dt}\left(e^{-\zeta\omega_n t} \sin \omega_d t\right)$$

$$= \frac{(I - i_{L0})}{\omega_n \sqrt{1-\zeta^2}} e^{-\zeta\omega_n t} \left[-\zeta\omega_n \sin \omega_d t + \omega_n\sqrt{1-\zeta^2} \cos \omega_d t\right]$$

$$i_C(t) = \frac{I - i_{L0}}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t}\left[-\zeta \sin \omega_d t + \sqrt{1-\zeta^2} \cos \omega_d t\right] \quad \text{---} \quad (**)$$

$$i_C(t) = -\frac{I - i_{L0}}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t} \sin(\omega_d t - \phi), \quad \phi = \tan^{-1}\frac{\sqrt{1-\zeta^2}}{\zeta}$$

$$i_L(t) = I - i_R - i_C$$

$$= I - \frac{(I - i_{L0})}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t}\left[2\zeta \sin\omega_d t - \zeta \sin \omega_d t + \sqrt{1-\zeta^2} \cos \omega_d t\right]$$

$$= I - \frac{(I - i_{L0})}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t}\left[\zeta \sin\omega_d t + \sqrt{1-\zeta^2} \cos \omega_d t\right]$$

$$i_L(t) = I - \frac{(I - i_{L0})}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t} \sin(\omega_d t + \phi)$$

$$i_{L0} = 0$$

$$i_L(t) = I\left[1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}} \cdot \sin(\omega_d t + \phi)\right]$$

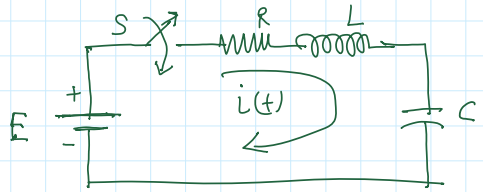
$i_L(0) = 0$

$i_L(\infty) = I$

Comparison summary of RLC series and RLC Parallel Circuit's Transient Response

RLC- Series Circuit	RLC- Parallel Circuit

RLC- Series Circuit



$$i(t) = i_L(t) = \frac{(E - V_{C0})}{\omega_d L} e^{-\zeta\omega_n t} \cdot \sin \omega_d t$$

When $V_{C0} = 0$

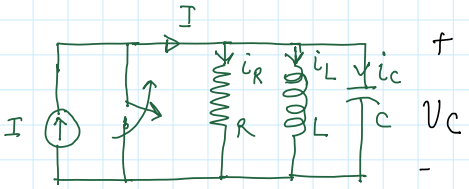
$$i(t) = i_L(t) = \frac{E}{\omega_d L} e^{-\zeta\omega_n t} \sin \omega_d t$$

$$V_R(t) = E * \frac{2\zeta}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t} \sin \omega_d t$$

$$V_L(t) = -\frac{E}{\sqrt{1-\zeta^2}} \cdot e^{-\zeta\omega_n t} \sin(\omega_d t - \phi)$$

$$V_C(t) = E\left\{1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}} \cdot \sin(\omega_d t + \phi)\right\}$$

RLC- Parallel Circuit



$$V(t) = V_C(t) = \frac{(I - i_{L0})}{\omega_d C} e^{-\zeta\omega_n t} \sin \omega_d t$$

When $i_{L0} = 0$

$$V(t) = V_C(t) = \frac{I}{\omega_d C} e^{-\zeta\omega_n t} \cdot \sin \omega_d t$$

$$i_R(t) = I * \frac{2\zeta}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t} \cdot \sin \omega_d t$$

$$i_C(t) = -\frac{I}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t} \sin(\omega_d t - \phi)$$

$$i_L(t) = I\left\{1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}} \cdot \sin(\omega_d t + \phi)\right\}$$

$$\phi = \tan^{-1}\frac{\sqrt{1-\zeta^2}}{\zeta} \; ; \; \omega_d = \omega_n\sqrt{1-\zeta^2} \; , \; \omega_n = \sqrt{\frac{1}{LC}}$$

$$\frac{R}{L} = 2\zeta\omega_n \text{ (RLC Series)} \; ; \; \frac{1}{RC} = 2\zeta\omega_n \text{ RLC-(parallel)}$$